

EVIDENCIAS – PROYECTO 1 SISTEMAS OPERATIVOS 1

CREACION 3 APIs

```
salojmmp@salojmmp-VirtualBox:~/202503517_LAB_S01_252026/api1$ cat>main.go<<'EOF'
>
package main

import (
    "encoding/json"
    "fmt"
    "log"
    "net/http"
    "time"
)

const PORT = "8001"
const CARNET = "202503517"
const VM = "VM1"

func healthHandler(w http.ResponseWriter, r *http.Request) {
    response := map[string]interface{}{
        "status":    "UP",
        "message":   "API1 is Ready",
        "timestamp": time.Now().Format(time.RFC3339),
        "VM":        VM,
        "carnet":    CARNET,
    }
    w.Header().Set("Content-Type", "application/json")
    json.NewEncoder(w).Encode(response)
}

func main() {
    http.HandleFunc("/health", healthHandler)
    fmt.Printf("API1 running on http://localhost:%s\n", PORT)
    log.Fatal(http.ListenAndServe(":"+PORT, nil))
}
EOF
salojmmp@salojmmp-VirtualBox:~/202503517_LAB_S01_252026/api1$
```

Q Buscar historial

```
salojmmp@salojmmp-VirtualBox:~$ ls
202503517_LAB_S01_2S2026  Documentos  Imágenes  Plantillas  Videos  go1.23.0.linux-amd64.tar.gz
Descargas                Escritorio  Música    Público     go        snap
salojmmp@salojmmp-VirtualBox:~$ cd 202503517_LAB_S01_2S2026
salojmmp@salojmmp-VirtualBox:~/202503517_LAB_S01_2S2026$ ls
README.md  api1  api2  api3  docs
salojmmp@salojmmp-VirtualBox:~/202503517_LAB_S01_2S2026$ cd api2
salojmmp@salojmmp-VirtualBox:~/202503517_LAB_S01_2S2026/api2$ cat >main.go<< 'EOF'
> package main
> import (
>     "encoding/json"
>     "fmt"
>     "log"
>     "net/http"
>     "time"
> )
> const PORT="8002"
> const CARNET="202503517"
> const VM="VM1"
> func healthHandler(w http.ResponseWriter, r *http.Request) {
>     response := map[string]interface{}{
>         "status":    "UP",
>         "message":    "API2 is Ready",
>         "timestamp":  time.Now().Format(time.RFC3339),
>         "VM":         VM,
>         "carnet":     CARNET,
>     }
>     w.Header().Set("Content-Type", "application/json")
>     json.NewEncoder(w).Encode(response)
> }
>
> func main() {
>     http.HandleFunc("/health", healthHandler)
>     fmt.Printf("API2 running on http://localhost:%s\n", PORT)
>     log.Fatal(http.ListenAndServe(":"+PORT, nil))
> }
EOF
salojmmp@salojmmp-VirtualBox:~/202503517_LAB_S01_2S2026/api2$ go run main.go
API2 running on http://localhost:8002
^Csignal: interrupt
salojmmp@salojmmp-VirtualBox:~/202503517_LAB_S01_2S2026/api2$
```

Q Buscar historial

```
salojmmp@salojmmp-VirtualBox: ~/202503517_LAB_S01_2S2026/api3 — go run main.go
~/202503517_LAB_S01_2S2026/api3

salojmmp@salojmmp-VirtualBox:~$ cd 202503517_LAB_S01_2S2026
salojmmp@salojmmp-VirtualBox:~/202503517_LAB_S01_2S2026$ cd api3
salojmmp@salojmmp-VirtualBox:~/202503517_LAB_S01_2S2026/api3$ cat>main.go<<'EOF'
package main
import (
    "encoding/json"
    "fmt"
    "log"
    "net/http"
    "time"
)

const PORT = "8003"
const CARNET = "202503517"
const VM = "VM2"

func healthHandler(w http.ResponseWriter, r *http.Request) {
    response := map[string]interface{}{
        "status":    "UP",
        "message":    "API3 is Ready",
        "timestamp": time.Now().Format(time.RFC3339),
        "VM":        VM,
        "carnet":    CARNET,
    }
    w.Header().Set("Content-Type", "application/json")
    json.NewEncoder(w).Encode(response)
}

func main() {
    http.HandleFunc("/health", healthHandler)
    fmt.Printf("API3 running on http://localhost:%s\n", PORT)
    log.Fatal(http.ListenAndServe(":"+PORT, nil))
}
EOF
salojmmp@salojmmp-VirtualBox:~/202503517_LAB_S01_2S2026/api3$ go run main.go
API3 running on http://localhost:8003
```

CRACION DE ARCHIVO DE CONFIGURACION

```
salojmmp@salojmmp-VirtualBox:~$ cat > ~/202503517_LAB_S01_2S2026/api1/Dockerfile << 'EOF'
FROM golang:1.23-alpine

WORKDIR /app

COPY main.go .

RUN go build -o api1 main.go

EXPOSE 8001

CMD ["/api1"]
EOF
```

```
salojmmp@salojmmp-VirtualBox:~$ cat > ~/202503517_LAB_S01_2S2026/api2/Dockerfile << 'EOF'
FROM golang:1.23-alpine

WORKDIR /app

COPY main.go .

RUN go build -o api2 main.go

EXPOSE 8002

CMD ["/api2"]
EOF
salojmmp@salojmmp-VirtualBox:~$ cat > ~/202503517_LAB_S01_2S2026/api3/Dockerfile << 'EOF'
> FROM golang:1.23-alpine

WORKDIR /app

COPY main.go .

RUN go build -o api3 main.go

EXPOSE 8003

CMD ["/api3"]
EOF
```

CONSTRUCCION DE IMÁGENES

```
salojmmp@salojmmp-VirtualBox: ~/202503517_LAB_SO1_2S2026/api3 — -bash
salojmmp@salojmmp-VirtualBox:~$ cd ~/202503517_LAB_SO1_2S2026/api2 && docker build -t api2-202503517:latest .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 5.12kB
Step 1/6 : FROM golang:1.23-alpine
--> 383395b794df
Step 2/6 : WORKDIR /app
--> Using cache
--> d4d582298f15
Step 3/6 : COPY main.go .
--> 24278a4ecebe
Step 4/6 : RUN go build -o api2 main.go
--> Running in e3251702d329
--> Removed intermediate container e3251702d329
--> 8f882815d5ba
Step 5/6 : EXPOSE 8002
--> Running in d82e82f0e31d
--> Removed intermediate container d82e82f0e31d
--> b9a5db40526b
Step 6/6 : CMD ["/api2"]
--> Running in ce08ec339f56
--> Removed intermediate container ce08ec339f56
--> 86bcc1abe147
Successfully built 86bcc1abe147
Successfully tagged api2-202503517:latest
salojmmp@salojmmp-VirtualBox:~/202503517_LAB_SO1_2S2026/api2$ cd ~/202503517_LAB_SO1_2S2026/api3 && docker build -t
02503517:latest .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 5.12kB
Step 1/6 : FROM golang:1.23-alpine
--> 383395b794df
Step 2/6 : WORKDIR /app
--> Using cache
--> d4d582298f15
Step 3/6 : COPY main.go .
--> 9a5edb11c535
Step 4/6 : RUN go build -o api3 main.go
--> Running in a72b5fb828bb
--> Removed intermediate container a72b5fb828bb
--> 8324fb4d48bb
Step 5/6 : EXPOSE 8003
--> Running in 5c7177cf4293

salojmmp@salojmmp-VirtualBox:~$ docker images | grep 202503517
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
api1-202503517:latest c00f65b3419d 481MB 101MB
api2-202503517:latest 86bcc1abe147 481MB 101MB
api3-202503517:latest 33d9092ae661 481MB 101MB
```

CREACION DE LAS MVs

```
salojmmp@salojmmp-VirtualBox:~$ virt-install \
--name vm1-containerd \
--memory 2048 \
--vcpus 2 \
--disk size=15,format=qcow2 \
--cdrom /var/lib/libvirt/images/ubuntu-22.04.5-live-server-amd64.iso \
--os-variant ubuntu22.04 \
--network default \
--graphics vnc,listen=127.0.0.1 \
--noautoconsole

Empezando la instalación...
Asignando 'vm1-containerd.qcow2' | 15 GB 00:00:00
Creando el dominio... | 00:00:00

El dominio todavía se está ejecutando. Puede que la instalación esté en curso.
Puede reconectarse a la consola para completar el proceso de instalación.
```


SE INSTALA CONTAINERD

```
ubuntu@vm1-containerd:~$ sudo apt install -y containerd
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Se instalarán los siguientes paquetes adicionales:
  runc
Se instalarán los siguientes paquetes NUEVOS:
  containerd runc
0 actualizados, 2 nuevos se instalarán, 0 para eliminar y 59 no actualizados.
Se necesita descargar 37,8 MB de archivos.
Se utilizarán 140 MB de espacio de disco adicional después de esta operación.
Des:1 http://gt.archive.ubuntu.com/ubuntu jammy-updates/main amd64 runc amd64 1.3.4-0ubuntu1~22.04.1 [9.569 kB]
Des:2 http://gt.archive.ubuntu.com/ubuntu jammy-updates/main amd64 containerd amd64 2.2.1-0ubuntu1~22.04.2 [28,3 MB]
Descargados 37,8 MB en 4s (9.452 kB/s)
Seleccionando el paquete runc previamente no seleccionado.
(Leyendo la base de datos ... 75402 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../runc_1.3.4-0ubuntu1~22.04.1_amd64.deb ...
Desempaquetando runc (1.3.4-0ubuntu1~22.04.1) ...
Seleccionando el paquete containerd previamente no seleccionado.
Preparando para desempaquetar .../containerd_2.2.1-0ubuntu1~22.04.2_amd64.deb ...
Desempaquetando containerd (2.2.1-0ubuntu1~22.04.2) ...
Configurando runc (1.3.4-0ubuntu1~22.04.1) ...
Configurando containerd (2.2.1-0ubuntu1~22.04.2) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /lib/systemd/system/containerd.service.
Procesando disparadores para man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@vm1-containerd:~$ ubuntu@vm1-containerd:~$ sudo apt install -y containerd

ubuntu@vm1-containerd:~$ sudo systemctl start containerd
ubuntu@vm1-containerd:~$ sudo systemctl enable containerd
ubuntu@vm1-containerd:~$ containerd --version
containerd github.com/containerd/containerd/v2.2.1
```

SE INSTALA PODMAN

```
Configurando podman (3.4.4+ds1-1ubuntu1.22.04.3) ...
Created symlink /etc/systemd/user/default.target.wants/podman.service → /usr/lib/systemd/user/podman.service.
Created symlink /etc/systemd/user/sockets.target.wants/podman.socket → /usr/lib/systemd/user/podman.socket.
Created symlink /etc/systemd/system/default.target.wants/podman.service → /lib/systemd/system/podman.service.
Created symlink /etc/systemd/system/sockets.target.wants/podman.socket → /lib/systemd/system/podman.socket.
Created symlink /etc/systemd/system/default.target.wants/podman-auto-update.service → /lib/systemd/system/podman-auto-update.service.
Created symlink /etc/systemd/system/timers.target.wants/podman-auto-update.timer → /lib/systemd/system/podman-auto-update.timer.
Created symlink /etc/systemd/system/default.target.wants/podman-restart.service → /lib/systemd/system/podman-restart.service.
Configurando dconf-service (0.40.0-3ubuntu0.1) ...
Configurando libavahi-glib1:amd64 (0.8-5ubuntu5.5) ...
Configurando libavahi-client3:amd64 (0.8-5ubuntu5.5) ...
Configurando dconf-gsettings-backend:amd64 (0.40.0-3ubuntu0.1) ...
Configurando gsettings-desktop-schemas (42.0-1ubuntu1) ...
Procesando disparadores para man-db (2.10.2-1) ...
Procesando disparadores para dbus (1.12.20-2ubuntu4.1) ...
Procesando disparadores para libglib2.0-0:amd64 (2.72.4-0ubuntu2.9) ...
Procesando disparadores para libc-bin (2.35-0ubuntu3.14) ...
Configurando glib-networking:amd64 (2.72.0-1) ...
Configurando libsoup2.4-1:amd64 (2.74.2-3ubuntu0.7) ...
Configurando libostree-1-1:amd64 (2022.2-3) ...
Configurando buildah (1.23.1+ds1-2) ...
Procesando disparadores para libc-bin (2.35-0ubuntu3.14) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@vm1-containerd:~$ sudo systemctl start podman
ubuntu@vm1-containerd:~$ sudo systemctl enable podman
ubuntu@vm1-containerd:~$ podman --version
podman version 3.4.4
```

SE INSTALA DOCKER


```
Seleccionando el paquete docker.io previamente no seleccionado.
Preparando para desempaquetar .../6-docker.io_29.1.3-0ubuntu3~22.04.2_amd64.deb ...
Desempaquetando docker.io (29.1.3-0ubuntu3~22.04.2) ...
Seleccionando el paquete ubuntu-fan previamente no seleccionado.
Preparando para desempaquetar .../7-ubuntu-fan_0.12.16_all.deb ...
Desempaquetando ubuntu-fan (0.12.16) ...
Configurando dnsmasq-base (2.90-0ubuntu0.22.04.4) ...
Configurando runc (1.3.4-0ubuntu1~22.04.1) ...
Configurando dns-root-data (2024071801~ubuntu0.22.04.1) ...
Configurando bridge-utils (1.7-1ubuntu3) ...
Configurando pigz (2.6-1) ...
Configurando containerd (2.2.1-0ubuntu1~22.04.2) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /lib/systemd/system/containerd.service.
Configurando ubuntu-fan (0.12.16) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ubuntu-fan.service → /lib/systemd/system/ubuntu-fan.service.
Configurando docker.io (29.1.3-0ubuntu3~22.04.2) ...
Adding group 'docker' (GID 119) ...
Done.
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /lib/systemd/system/docker.socket.
Procesando disparadores para dbus (1.12.20-2ubuntu4.1) ...
Procesando disparadores para man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@vm1-containerd:~$ sudo systemctl start docker
sudo systemctl enable docker
ubuntu@vm1-containerd:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3~22.04.2
```

SE INSATALA ZOT

```

t=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiaXVkiJjoicmVsZWZzZS1hc3NldHMuz2L0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc4NzI3NTUxMywibmJmIjoxNzg3Mjc0TEZLCjwYXRoIjoicmVsZWZzZWZzc2V0CHJvZHVjdGlubi5ibG9iLmNvcmlud2luZC93cy5uZXQifQ.rvPXqaIPGOXP1KjKuBECET2jlv1TYt8mK9h8ctvTwyY&response-content-disposition=attachment%3B%20filename%3Dzot-linux-amd64&response-content-type=application%2Foctet-stream [following]
--2026-08-21 00:25:13-- https://release-assets.githubusercontent.com/github-production-release-asset/193163140/244527a0-dfc4-4185-938a-1e06bf7aa4f1?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-08-21T01%3A00%3A37Z&rscd=attachment%3B+filename%3Dzot-linux-amd64&rscd=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-08-21T00%3A00%3A17Z&ske=2026-08-21T01%3A00%3A37Z&sks=b&skv=2018-11-09&sig=lZ7yI32GTtcQprvbZntL%2FN2zc0RBjIg2T%2FQMHkZUG50%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiaXVkiJjoicmVsZWZzZS1hc3NldHMuz2L0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc4NzI3NTUxMywibmJmIjoxNzg3Mjc0TEZLCjwYXRoIjoicmVsZWZzZWZzc2V0CHJvZHVjdGlubi5ibG9iLmNvcmlud2luZC93cy5uZXQifQ.rvPXqaIPGOXP1KjKuBECET2jlv1TYt8mK9h8ctvTwyY&response-content-disposition=attachment%3B%20filename%3Dzot-linux-amd64&response-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.111.133, 185.199.108.133, 185.199.109.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 148185088 (141M) [application/octet-stream]
Saving to: 'zot-linux-amd64'

zot-linux-amd64          100%[=====] 141,32M  5,20MB/s   in 48s

2026-08-21 00:26:02 (2,93 MB/s) - 'zot-linux-amd64' saved [148185088/148185088]

ubuntu@vm1-containerd:/tmp$ sudo mkdir -p /var/lib/zot
ubuntu@vm1-containerd:/tmp$ sudo chown ubuntu:ubuntu /var/lib/zot
ubuntu@vm1-containerd:/tmp$ cat > ~/zot-config.json << 'EOF'
> {
  "distSpecVersion": "1.1.0",
  "storage": {
    "rootDirectory": "/var/lib/zot"
  },
  "http": {
    "address": "0.0.0.0",
    "port": "5000"
  },
  "log": {
    "level": "debug"
  }
}
EOF
ubuntu@vm1-containerd:/tmp$ sudo /tmp/zot-linux-amd64 serve ~/zot-config.json > ~/zot.log 2>&1 &
[1] 2040
ubuntu@vm1-containerd:/tmp$ curl http://localhost:5000/v2/

ubuntu@vm1-containerd:/tmp$ curl -i http://localhost:5000/v2/
HTTP/1.1 200 OK
Access-Control-Allow-Headers: Authorization,content-type,X-ZOT-API-CLIENT
Access-Control-Allow-Methods: GET,OPTIONS
Access-Control-Allow-Origin: *
Content-Type: application/json
Docker-Distribution-API-Version: registry/2.0
WWW-Authenticate: basic realm=
Date: Fri, 21 Aug 2026 00:44:43 GMT
Content-Length: 0

```

SE EXPORTAN LAS IMÁGENES

```

salojmmp@salojmmp-VirtualBox:~$ docker save api1-202503517:latest > /tmp/api1-202503517.tar
salojmmp@salojmmp-VirtualBox:~$ docker save api2-202503517:latest > /tmp/api2-202503517.tar
salojmmp@salojmmp-VirtualBox:~$ docker save api3-202503517:latest > /tmp/api3-202503517.tar
salojmmp@salojmmp-VirtualBox:~$ virsh domifaddr vm1-containerd

```

SE COPIAN A LA MV1

```
salojmmp@salojmmp-VirtualBox:~$ scp /tmp/api1-202503517.tar ubuntu@192.168.122.80:/tmp/
ubuntu@192.168.122.80's password:
api1-202503517.tar 100% 96MB 12.4MB/s 00:07
salojmmp@salojmmp-VirtualBox:~$ scp /tmp/api2-202503517.tar ubuntu@192.168.122.80:/tmp/
ubuntu@192.168.122.80's password:
api2-202503517.tar 100% 96MB 11.7MB/s 00:08
salojmmp@salojmmp-VirtualBox:~$ ssh ubuntu@192.168.122.80
ubuntu@192.168.122.80's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-190-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of vie 21 ago 2026 01:11:39 UTC

System load: 0.01          Processes:              128
Usage of /:  51.8% of 9.75GB Users logged in:             0
Memory usage: 11%         IPv4 address for enp1s0: 192.168.122.80
Swap usage:  0%

El mantenimiento de seguridad expandido para Applications está desactivado

Se pueden aplicar 67 actualizaciones de forma inmediata.
3 de estas son actualizaciones de seguridad estándares.
Para ver estas actualizaciones adicionales, ejecute: apt list --upgradable

1 actualización de seguridad adicional se puede aplicar con ESM Apps.
Aprenda más sobre cómo activar el servicio ESM Apps at https://ubuntu.com/esm

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Fri Aug 21 01:09:37 2026 from 192.168.122.1
ubuntu@vm1-containerd:~$ sudo ctr -n k8s.io images import /tmp/api1-202503517.tar
[sudo] password for ubuntu:
docker.io/library/api1 202503517:latest          saved
application/vnd.oci.image.manifest.v1+json sha256:c00f65b3419df012fc653db1095a82879f572f65ab4397e0baeae1bd37ef981d
Importing          elapsed: 19.7s total:  0.0 B (0.0 B/s)
ubuntu@vm1-containerd:~$ sudo ctr -n k8s.io images import /tmp/api2-202503517.tar
docker.io/library/api2 202503517:latest          saved
application/vnd.oci.image.manifest.v1+json sha256:86bcc1abe147c38188f9fa269b459aa73ec995d47b2a9a9bee80dbb024e0f6ff
Importing          elapsed: 4.6 s total:  0.0 B (0.0 B/s)
```


SE COPIA A MV2

```
salojrmp@salojrmp-VirtualBox:~$ scp /tmp/api3-202503517.tar ubuntu@192.168.122.80:/tmp/
The authenticity of host '192.168.122.80 (192.168.122.80)' can't be established.
ED25519 key fingerprint is: SHA256:foc3qnNrW0m5pRAIZQft22rCQ0GMfZUhJ+708gJPpyM
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.122.80' (ED25519) to the list of known hosts.
ubuntu@192.168.122.80's password:
api3-202503517.tar                                                                 100% 96MB 45.3MB/s 00:02
salojrmp@salojrmp-VirtualBox:~$ ssh ubuntu@192.168.122.80
ubuntu@192.168.122.80's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-190-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of vie 21 ago 2026 01:18:07 UTC

System load: 0.18           Processes:              136
Usage of /:  51.5% of 9.75GB Users logged in:          0
Memory usage: 11%          IPv4 address for enp1s0: 192.168.122.80
Swap usage:  0%

El mantenimiento de seguridad expandido para Applications está desactivado

Se pueden aplicar 67 actualizaciones de forma inmediata.
3 de estas son actualizaciones de seguridad estándares.
Para ver estas actualizaciones adicionales, ejecute: apt list --upgradable

1 actualización de seguridad adicional se puede aplicar con ESM Apps.
Aprenda más sobre cómo activar el servicio ESM Apps at https://ubuntu.com/esm

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Aug 20 02:10:59 2026 from 192.168.122.1
ubuntu@vm1-containerd:~$ podman load -i /tmp/api3-202503517.tar
Getting image source signatures
Copying blob 9824c27679d3 done
Copying blob 8371a51cbc44 done
Copying blob d5791340ef18 done
Copying blob d3178a7b2709 done
Copying blob 4f4fb700ef54 done
Copying blob b815907a5e0f done
```

SE COPIA A MV3

```
salojmmp@salojmmp-VirtualBox:~$ # 1. Limpiamos el candado de SSH
ssh-keygen -f '/home/salojmmp/.ssh/known_hosts' -R '192.168.122.80'

# 2. Enviamos las 3 APIs de un solo viaje por SCP
scp /tmp/api1-202503517.tar /tmp/api2-202503517.tar /tmp/api3-202503517.tar ubuntu@192.168.122.80:/tmp/
# Host 192.168.122.80 found: line 1
# Host 192.168.122.80 found: line 2
# Host 192.168.122.80 found: line 3
/home/salojmmp/.ssh/known_hosts updated.
Original contents retained as /home/salojmmp/.ssh/known_hosts.old
The authenticity of host '192.168.122.80 (192.168.122.80)' can't be established.
ED25519 key fingerprint is: SHA256:AhLVlnX/ZqNedtjr3VpzvtpbBiZUT02zDpeRcCsTbzY
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.122.80' (ED25519) to the list of known hosts.
ubuntu@192.168.122.80's password:
api1-202503517.tar                                100% 96MB 33.8MB/s  00:02
api2-202503517.tar                                100% 96MB 26.5MB/s  00:03
api3-202503517.tar                                100% 96MB 20.8MB/s  00:04

ubuntu@vm1-container:~$ sudo docker load -i /tmp/api1-202503517.tar
[sudo] password for ubuntu:
Loaded image: api1-202503517:latest
ubuntu@vm1-container:~$ sudo docker load -i /tmp/api2-202503517.tar
Loaded image: api2-202503517:latest
ubuntu@vm1-container:~$ sudo docker load -i /tmp/api3-202503517.tar
Loaded image: api3-202503517:latest
ubuntu@vm1-container:~$ sudo docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
api1-202503517:latest	c00f65b3419d	481MB	101MB	
api2-202503517:latest	86bcc1abe147	481MB	101MB	
api3-202503517:latest	33d9092ae661	481MB	101MB	

API 3 corre en mv2

```
ubuntu@vm1-container:~$ podman run -d --net host localhost/latest:latest
10b4ec38f86cb663bdd21b370d2ba5a74935a29985e254867536d5d69a5665c1
ubuntu@vm1-container:~$ podman ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
10b4ec38f86c   localhost/latest:latest             ./api3                  12 seconds ago Up 12 seconds 8003         blissful_knuth
ubuntu@vm1-container:~$ curl http://localhost:8003/health
{"VM": "VM2", "carnet": "202503517", "message": "API3 is Ready", "status": "UP", "timestamp": "2026-08-21T01:48:06Z"}
ubuntu@vm1-container:~$ exit
logout
Connection to 192.168.122.80 closed.
```

LAS 2 APIs CORREN EN MV3

```
ubuntu@vm1-container:~$ sudo docker run -d --network host api1-202503517:latest /app/api1
[sudo] password for ubuntu:
dfcfe759ad687c1fe3c5758c091b26ea7173a455b0e72f121aae3a186e8f4137
ubuntu@vm1-container:~$ sudo docker run -d --network host api2-202503517:latest /app/api2
eff390e0e6ab1a17b88c86dc6402d5a9ffcebeed85f010301ffce45c29243e90
ubuntu@vm1-container:~$ sudo docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
eff390e0e6ab   api2-202503517:latest             "/app/api2"             7 seconds ago Up 6 seconds 8002         relaxed_hugle
dfcfe759ad68   api1-202503517:latest             "/app/api1"             10 seconds ago Up 9 seconds 8001         nice_germain
ubuntu@vm1-container:~$ curl http://localhost:8001/health
curl http://localhost:8002/health
{"VM": "VM1", "carnet": "202503517", "message": "API1 is Ready", "status": "UP", "timestamp": "2026-08-21T01:52:50Z"}
{"VM": "VM1", "carnet": "202503517", "message": "API2 is Ready", "status": "UP", "timestamp": "2026-08-21T01:52:50Z"}
```


LA COMUNICACION ENTRE MVs ES EXITOSA

```
salojmmp@salojmmp-VirtualBox:~$ docker run -d --name api1 --network host api1-202503517:latest
docker run -d --name api2 --network host api2-202503517:latest
docker run -d --name api3 --network host api3-202503517:latest
eb120cbae59ff52621cab44705f1430294b62bd10c6dc04ff8b68e20ab04a213
2d91eb989724351aa1c7c7a99a012d83816b780334cd80a75970cd3168a0a3be
047fa01609a5d2f31499c662f5f52450aafe1a89456d86358a6f55752420a256
salojmmp@salojmmp-VirtualBox:~$ curl http://localhost:8001/api1/202503517/call-api2
{"apiname":"API2","carnet":"202503517","connection":true,"message":"The API2 located on the VM1 is working"}
salojmmp@salojmmp-VirtualBox:~$ curl http://localhost:8001/api1/202503517/call-api3

{"apiname":"API3","carnet":"202503517","connection":false,"message":"ERROR: The API3 located on the VM is not working"}
salojmmp@salojmmp-VirtualBox:~$
salojmmp@salojmmp-VirtualBox:~$ curl http://localhost:8002/api2/202503517/call-api3

{"apiname":"API3","carnet":"202503517","connection":false,"message":"ERROR: The API3 located on the VM is not working"}
```